

Lecture 3: Normal Linear Models — Notation and R Fitting

MATH3701 Statistical Modelling

Richard P Mann

MATH3701 Statistical Modelling

Course notes: Chapter 2, Sections 2.5–2.7

MATH3701 Statistical Modelling notes, Chapter 2

The $\sim$ operator: “is modeled by”

Symbol	Meaning
$y \sim x$	y is modeled by x
1	Intercept (included by default)
E	Main effect of variable E
$E + F$	Main effects of E and F
$E : F$	Interaction between E and F
$E * F$	Main effects + interaction ($= E + F + E : F$)
-E	Remove term E from model
-1 or 0	Remove intercept

More Formula Notation

Symbol	Meaning
E / F	Nested factors: $E + E : F$
$\text{poly}(E, \ell)$	Polynomial of degree ℓ in E
$I()$	“As is” — use arithmetic literally
$.$	All other variables in data frame

Examples:

- $y \sim x + I(x^2)$: quadratic regression
- $y \sim \text{poly}(x, 3)$: cubic polynomial (orthogonal)
- $y \sim . - z$: all variables except z

Formula Examples

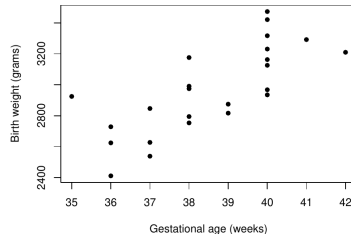
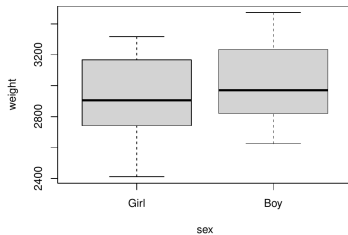
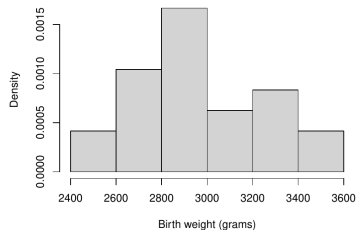
Birth weight example:

- `weight ~ age`: simple regression on age
- `weight ~ age + sex`: parallel lines model
- `weight ~ age * sex`: separate lines (with interaction)
- `weight ~ age + sex + age:sex`: same as above

More examples:

- `yield ~ variety + poly(rainfall, 2)`: ANCOVA with quadratic term
- `score ~ school/class`: students nested in classes within schools
- `profit ~ investment - 1`: regression through the origin

The Birthweight Data



Both age and sex look like useful predictors — exactly what `model2 <- lm(weight ~ age + sex)` will fit.

The `lm()` Function

Basic syntax:

```
my.lm <- lm(formula, data = mydata)
```

Example:

```
# Fit the parallel lines model
model2 <- lm(weight ~ age + sex, data = birthwt)

# View results
summary(model2)
```

Note: R automatically handles:

- Design matrix construction
- Dummy coding for factors
- Parameter estimation

Extracting Information from lm Objects

```
# Coefficient estimates
coefficients(my.lm)    # or coef(my.lm)

# Fitted values
fitted.values(my.lm)   # or fitted(my.lm)

# Residuals
residuals(my.lm)       # or resid(my.lm)

# Residual degrees of freedom
df.residual(my.lm)

# Design matrix
model.matrix(my.lm)
```

The summary() Output

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	-1773.324	820.906	-2.160	0.0423 *
age	120.894	21.423	5.643	1.26e-05 ***
sex	163.039	72.814	2.239	0.0360 *

Residual standard error: 172.2 on 21 degrees of freedom

Multiple R-squared: 0.6387, Adjusted R-squared: 0.6043

F-statistic: 18.56 on 2 and 21 DF, p-value: 2.237e-05

Key elements:

- **Estimates:** $\hat{\beta}$ values
- **Std. Error:** $\sqrt{\text{diag}(\hat{\sigma}^2(\mathbf{X}'\mathbf{X})^{-1})}$
- **t value:** Estimate / Std. Error
- **p-value:** Test of $H_0 : \beta_j = 0$

The `anova()` Function

Single model: Sequential analysis of variance

```
anova(my.lm)
```

Multiple models: Compare nested models

```
anova(model1, model2, model3)
```

Output includes:

- Sum of squares for each term
- Degrees of freedom
- Mean squares
- F -statistics and p -values

Making Predictions

Predict at new values of the explanatory variables:

```
# Create new data
new_data <- data.frame(age = c(38, 40, 42),
                      sex = c(0, 1, 1))

# Point predictions
predict(model2, newdata = new_data)

# With confidence intervals
predict(model2, newdata = new_data,
       interval = "confidence")

# With prediction intervals
predict(model2, newdata = new_data,
       interval = "prediction")
```

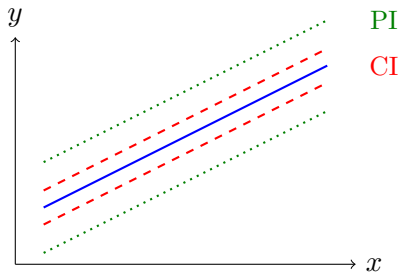
Confidence vs. Prediction Intervals

Confidence interval for $\mathbb{E}[Y \mid \mathbf{x}]$:

- Uncertainty about the *mean* response
- Narrower

Prediction interval for new Y :

- Uncertainty about an *individual* observation
- Wider (includes σ^2)

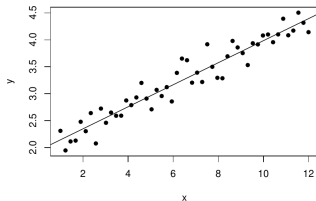


Key diagnostic plots:

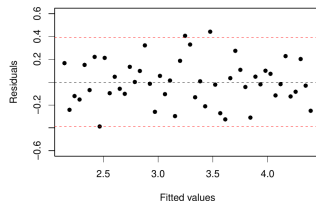
- ➊ **Residuals vs. fitted:** Check linearity, constant variance
- ➋ **Q-Q plot:** Check normality of residuals
- ➌ **Scale-location:** Check homoscedasticity
- ➍ **Residuals vs. leverage:** Identify influential points

```
# All four plots  
par(mfrow = c(2, 2))  
plot(my.lm)
```

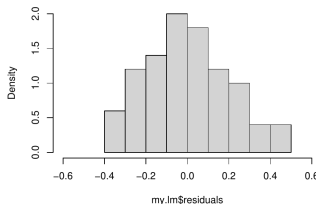
Example: Diagnostics on a Toy Dataset



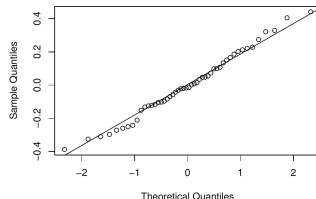
(a) Data and fitted line



(b) Residuals vs. fitted



(c) Histogram of residuals



(d) Normal Q-Q plot

A well-behaved fit: residuals scatter randomly, histogram and Q-Q plot are consistent with normality.

What to Look For

Residuals vs. Fitted:

- **Good:** Random scatter around 0
- **Bad:** Curved pattern $\Rightarrow$ nonlinearity
- **Bad:** Funnel shape $\Rightarrow$ heteroscedasticity

Q-Q Plot:

- **Good:** Points follow diagonal line
- **Bad:** S-shape $\Rightarrow$ heavy/light tails
- **Bad:** Curve $\Rightarrow$ skewness

If assumptions violated:

- Transform response (log, sqrt)
- Add polynomial terms
- Use a different model (e.g., GLM)

Declaring Factors in R

Problem: R may treat numeric codes as continuous.

```
# Sex coded as 0/1
sex <- c(0, 0, 1, 1, 0, 1)

# Wrong: treated as quantitative
lm(y ~ sex) # Only one coefficient for sex

# Right: declare as factor
sex <- as.factor(sex)
lm(y ~ sex) # Now categorical
```

Best practice: Always use `factor()` or meaningful labels:

```
sex <- factor(c("F", "F", "M", "M", "F", "M"))
```

Key points:

- R formula notation provides concise model specification
- `lm()` fits normal linear models
- `summary()` gives coefficient estimates and tests
- `anova()` compares nested models
- `predict()` generates predictions with intervals
- Diagnostic plots check model assumptions
- Declare categorical variables as factors

Next lecture: Introduction to GLM theory — the exponential family.